

WASP-43b

R-2147

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_150404 · created 2026-08-12T00:47:39+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457116.77146 +/- 0.00092 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.197 +/- 0.057
Transit depth (Rp/Rs)^2	3.88 +/- 2.26 [%]
Orbital Inclination (inc)	85.68 +/- 2.9
Airmass coefficient 1 (a1)	0.2 +/- 0.12
Airmass coefficient 2 (a2)	1.01 +/- 0.59
Scatter in the residuals of the lightcurve fit is	60.89 %
Best Comparison Star	#2 - [557, 364]
Optimal Aperture	2.59
Optimal Annulus	7.88
Transit Duration (day)	0.0596 +/- 0.0061

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.197 $\pm$ 0.057 vs 0.1594 (deviation 0.47 σ)
Duration fit vs published	0.0596 d vs 0.0512 d (deviation 0.97 σ)
Mid-time O-C	2.0 min
Depth SNR	2.5
Residual scatter	60.89 %

Light curve

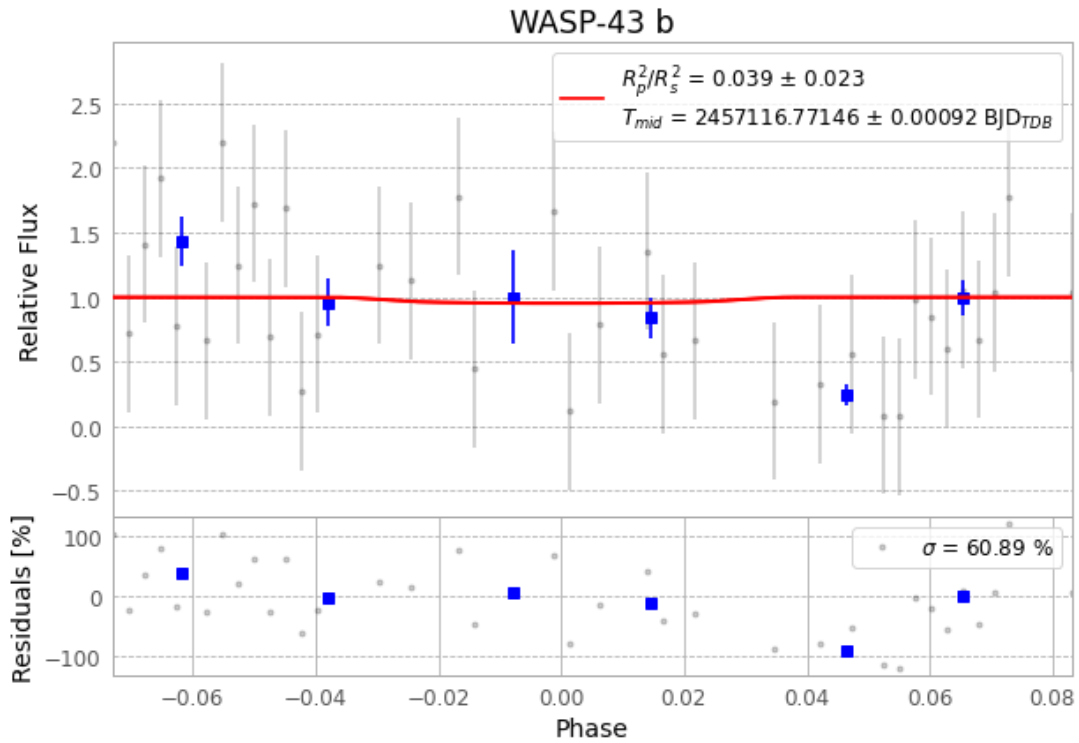


Figure 1 — FinalLightCurve_WASP-43 b_2015-04-03.png (SHA-256 a09a10e0b5261354...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [360, 127], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d77c2b6da95b9dfd...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

64 FITS frames (42 MB), WASP-43150404045712.FITS → WASP-43150404080612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-150404.log (3e9563e1cae2...), FinalLightCurve_WASP-43 b_2015-04-03.png (a09a10e0b526...), FinalLightCurve_WASP-43 b_2015-04-03.csv (69e78026c80d...)

Compute resources

Wall time 152.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-12 00:47Z.